Scientia Horticulturae 162 (2013) 39–42

Contents lists available at ScienceDirect

Scientia Horticulturae

journal homepage: www.elsevier.com/locate/scihorti

Short communication

Cape gooseberry (*Physalis peruviana* L.) phenology according to the BBCH phenological scale

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ARTICLE INFO

Article history:
Received 22 March 2013
Received in revised form 23 July 2013
Accepted 26 July 2013

Keywords:
Flowering
Fruit
Solanaceous
Bud
Colombia

ABSTRACT

The cape gooseberry (*Physalis peruviana* L.) is a shrub found native in the Andes that bears tomato-like fruit, which are sold commercially and exported from Colombia. We describe the BBCH-based phenological scale for this crop that was developed by observing three cape gooseberry ecotype accessions during a two-year study. The proposed BBCH phenological scale uses seven principal growth stages out of the nine principal stages identified for solanaceous fruits. The principal growth stages are (0) germination, (1) leaf development, (2) formation of side shoots, (5) inflorescence emergence, (6) flowering, (7) development of fruit and (8) ripening of fruit and seed. Early growth stages (0–1) in cape gooseberry are similar to the developmental stages of other solanaceous plants. The first bifurcation (stage 2) marks the beginning of floral development (stage 5). Thereafter, simultaneous developmental stages occur throughout the plant's phenology. It is common to observe early flowers, fully open flowers, immature and mature fruits on the same plant. The BBCH phenological scale developed for cape gooseberry is a useful tool for management and research practices and for subsequent investigations on any of the developmental stages the plant passes through.

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1. Introduction

Cape gooseberry (*Physalis peruviana* L.), a plant locally known as uchuva, is a herbaceous plant native to the Andes and the second export fruit of Colombia (Fischer et al., 2007). This plant has attracted great interest worldwide because it produces fruit that has medicinal, nutritious, and pharmaceutical properties. Health related components include antioxidants, polyunsaturated fatty acids, vitamins A, B, C, E and K1, phytosterols, and the presence of essential minerals (Puente et al., 2011).

The cape gooseberry belongs to the Solanaceae family. The genus *Physalis* includes about 100 species characterized by fruits that bear an inflated calyx (Legge, 1974). Whitson and Manos (2005) limited the genus to 75 to 90 species. Cape gooseberry is considered a perennial shrubby herb. In natural, non-managed conditions, it has been known to reach 1.0 to 1.5 m in height, with its typical sympodial branching (Fischer et al., 2011; Fischer and Miranda, 2012). The vegetative basal stem is somewhat woody toward its base, and after forming eight to twelve nodes, it branches twice dichasially so that usually four generative main stems are formed (Fischer, 2000). These can reach up to 2 m in height when plants are staked. The plant's high productivity in poor soils, ease of cultivation, low requirement for water and fertilizer and short production time make it an attractive potential crop (McCain, 1993; National Research Council, 1989).

Cape gooseberry growth is continuous. Both vegetative and reproductive meristems remain active during the plant's life cycle. Floral, vegetative and fruit growth and development occur simultaneously. Cape gooseberry can, thus, be classified as a plant with indeterminate growth habit (Fischer et al., 2011). Seed growth begins as a result of meristematic activity. The apical meristem elongates vegetatively forming lateral nodes with internode lengths averaging 5.0 cm (Fischer, 1995). Plant elongation continues as a single vegetative stem until it develops eight to 12 nodes (Fischer, 2000). The vegetative stage is characterized by the occurrence of simple, petiolated, alternate, heart-shaped leaves that are highly pubescent (Fischer, 2000). Leaf dimensions range from 5 to 15 cm long and 4 to 10 cm wide; however, under greenhouse conditions, leaves can reach up to 20 cm in length (Fischer, 1989). Each

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http://dx.doi.org/10.1016/j.scienta.2013.07.033

Table 1
Soil analysis.

Location	pH	N	Ca	K	Mg	Na	P	Cu	Fe	Zn	B
		Percent (%)	(meq/100 g)				(mg kg^{-1})				
Bogotá	6.0	0.40	10	0.51	4.2	4.98	45.5	1.05	432	19.3	0.91
Silvania	5.3	0.35	5.67	0.69	2.2	2.86	29.33	9.45	191.46	7.45	0.45
Fusagasugá	5.5	0.38	7.70	0.49	3.1	3.45	30.35	5.96	245.30	11.45	0.57

node on the main lower axis of the plant develops one leaf. Nodes on lateral reproductive branches generally develop two leaves (Fischer, 2000). The formation of eight to 12 nodes with internode lengths averaging 6.4 cm marks the beginning of the reproductive stage after the appearance of the first flower. Thus, each node bears one flower (reproductive bud), two petiolate leaves (vegetative buds) and, especially in the lower part of the plant, a lateral shoot (vegetative bud) (Fischer, 2000; Bernal, 1991). After the appearance of the first flower, two new shoots develop as bifurcated node outgrowths from the axillary buds at the base of the two leaves. These shoots elongate and form a node where another flower forms along with two leaves and two new shoots that elongate to form more reproductive structures (Fischer, 2000). First flowering of *Physalis* occurs between 70 and 80 days after seed germination (Gupta and Roy, 1981). The period from floral bud initiation to anthesis is 19 to 23 days (Gupta and Roy, 1981). The calyx (or husk), which is small at the beginning of fruit development, grows to a bladder-like, papery organ, which completely encloses the ripening fruit (Fischer et al., 2011). Fruits typically develop to about 2 cm diameter and turn from green to yellow upon ripening. Fruit harvest begins 4 to 7 months after seed germination, depending on the altitude (temperature) of the orchard site (Galvis et al., 2005).

Phenology is a useful tool to better understand plant life cycles. It has been studied in numerous herbaceous plants, such as rice, maize, sunflower, beans and peas among others (Meier, 1997). The extended BBCH-scale is a system for uniform encoding of phenologically similar growth stages of all mono- and dicotyledonous plant species (Meier, 1997). It's development was the product of a collaborative effort between the German Federal Biological Research Center for Agriculture and Forestry (BBA), the German Federal Office of Plant Varieties (BSA), the German Agrochemical Association (IVA) and the Institute for Vegetables and Ornamentals in Grossbeeren/Erfurt, Germany (IGZ) (Meier, 1997). According to Meier (1997), the entire developmental cycle of the plants is subdivided into ten clearly recognizable and distinguishable, sometimes long-lasting developmental phases. These principal growth stages are described using numbers from 0 to 9 in ascending order. Moreover, secondary stages are used if points of time or steps in plant development must be indicated precisely. In contrast to the principal growth stages secondary stages are defined as short, developmental steps characteristic of the respective plant species, which are passed successively during the respective principal growth stage. They are also coded by using the figures 0 to 9 (Meier, 1997). The two-digit code is a scale that offers the possibility of precisely defining all phenological growth stages for the majority of plant species (Meier, 1997). Within this context, no investigation using the BBCH scale has been developed for cape gooseberry. The aim of this work was to describe the phenological stages of cape gooseberry growth and development using the BBCH scale. This scale can be used by horticulturists and researchers to identify the development of the plant at a particular growth stage.

2. Materials and methods

2.1. Plant material

The phenological cycle of cape gooseberry was studied among tree accessions representing three ecotypes from Colombia, South Africa and Kenya. Seed-propagated plants of the Colombian ecotype were grown in Bogotá in outdoor conditions during two consecutive years, 2011 and 2012. Additional observations were made at various intervals between May and November, 2012 at orchards near Silvania and Fusagasugá in Cundinamarca State, Colombia. Observations of the phenological phases were recorded weekly from early seedling devolvement to fruit set. Plants planted in Bogotá were spaced 1 m × 2 m apart. Plants at Silvania and Fusagasugá were spaced 2 × 2.5 m between and among rows. All were irrigated and fertilized as required. South African and Kenyan ecotypes were observed in the field at Fusagasugá and Silvania. Soil conditions at Bogota, Fusagasugá and Silvania are characterized in Table 1.

2.2. Climatic conditions and elevation

Bogotá (4° 35′ 53″ N, 74° 4′ 33″ W) is located in Cundinamarca State, at 2600 m above sea level (m.a.s.l.) on the Eastern cordillera of the Andes. The climate of Bogotá is considered tropical cold. Mean monthly rainfall for the 2-year study was 76 mm, with an average relative humidity of 81%. Mean monthly temperature was a constant 13° ± C with a daily range of 13–17 °C. Mean monthly temperature in the orchard located near the village of Silvania (4°24′19″ N, 74°23′26″ W) was 20 °C± with a daily range of 20–24 °C at 1507 m.a.s.l., with annual precipitation of 1653 mm, and mean relative humidity of 80%. The mean monthly temperature conditions at Fusagasugá (4° 20′ 38″ North, 74° 22′ 4″ West), at an elevation of 1728 m.a.s.l., was 20° ± with a daily range of 20–23 °C with 1250 mm annual precipitation and 85% mean relative humidity.

2.3. Phenology

The proposed BBCH phenological scale for cape gooseberry is represented by seven principal growth stages out to the nine stages identified for solanaceous fruits (Meier, 1997). These principal stages begin with germination (0) and end with ripening of fruit and seed (8). We did not include the principal senescence stage (9) of solanaceous fruits, because this stage was not observed among plants due to timely harvest of the fruit. According to Meier (1997), secondary stages are used if steps in plant development must be indicated precisely. These are defined as short developmental steps in cape gooseberry. In the case of solanaceous fruits, it is necessary to further subdivide the secondary stage into more sub-stages. In this case, the three-digit code scale is used alongside the two-digit scale according to Meier (1997). The principal stages, (3) representing stem elongation or rosette growth, shoot development and stage (4), representing development of harvestable vegetative plant parts or vegetatively propagated organs (main shoot) were omitted from the scale of cape gooseberry because these do not apply.

3. Results and discussion

The phenological cycle identified for cape gooseberry (Fig. 1) was characterized as follows:

3.1 Principal growth stage 0: Germination (Secondary and sub stages listed in Table 2)



Fig. 1. Phenological stages of *Physalis peruviana* L. according to the BBCH scale. (A) Nine or more leaves on main shoot unfold, (B–D) first and second bifurcations apical shoots visible, note the first flower on bifurcation, (C, E) new shoot emerging, (F) floral buds, (G–H) first flower open, (I–J) fruit develop in clusters (normally each node in the reproductive plant part bears one fruit), (K) immature fruit, (L) mature fruit in colored calyx, (M) mature fruit, (N–O) simultaneous occurrence of flower, fruits and buds on the same plant.

Photos by Fernando Ramírez.

Table 2
Germination process.

2-digit/	3-digit code	Description
00	000	Dry seeds
00	001	Beginning of seed imbibition
03	003	Seed imbibition complete
05	005	Radicle emerges from seed
07	007	Hypocotyl with cotyledons breaking through seed coat
09	009	Emergence: cotyledons break through soil surface

Table 3
Leaf developmental process.

2-digit	3-digit code	Description
10	100	Cotyledons completely unfolded
11	101	First true leaf on main stem fully unfolded
12	102	Second leaf on main shoot unfolded
13	103	Third leaf on main shoot unfolded
1.	10.	Stages continuous till
19	109	Nine or more leaves on the main shoots unfold (Fig. 1A)

3.2 Principal growth stage 1: Leaf development (Secondary and sub stages listed in Table 3)

3.3 Principal growth stage 2: Formation of side shoots (This stage occurs simultaneously with stage five, inflorescence emergence) (Secondary and sub stages listed in Table 4)

3.4 Principal growth stage 3: Not applicable

3.5 Principal growth stage 4: Not applicable

3.6 Principal growth stage 5: Inflorescence emergence (Secondary and sub stages listed in Table 5)

Table 4
Side shoot formation process.

2-digit	3-digit code	Description
21	201	First bifurcation two apical shoots visible (Fig. 1B–D)
22	202	Second bifurcation two apical side shoots visible (Fig. 1B-D)
2.	20.	Stages continuous till
29	209	Nine or more apical bifurcated shoots visible

Table 5
Inflorescence emergence process.

2-digit	3-digit code	Description
51	501	First flower bud visible on first bifurcation (Fig. 1C)
52	502	Second flower bud visible on second bifurcation (Fig. 1D–E)
53	503	Third flower bud visible on third bifurcation
5.	50.	Stages continuous till
59	509	Nine or more flower buds visible on ninth bifurcation
–	51.	Stages continuous till
–	509	19th flower bud visible

Table 6
Flowering process.

2-digit	3-digit code	Description
61	601	First flower opens (Fig. 1G–H)
62	602	Second flower opens
63	603	Third flower opens
6.	60.	Stages continuous till
69	609	Nine or more flowers open
–	61.	Stages continuous till
–	609	19th flower opens

3.7 Principal growth stage 6: Flowering (This stage occurs simultaneously with stage seven, development of fruit) (Table 6)

3.8 Principal growth stage 7: Development of fruit (Table 7)

3.9 Principal growth stage 8: Ripening of fruit and seed (Table 8)

Early stages (0–1) in cape gooseberry are similar to those of other solanaceous plants such as tomato, and paprika (Feller et al.,

Table 7
Fruit developmental process.

2-digit	3-digit code	Description
71	701	First fruit reaches typical size and form (Fig. 1I, J)
72	702	Second fruit reaches typical size and form
73	703	Third fruit has reached typical size and form
7.	70.	Stages continuous till
79	709	Nine or more fruits have reached typical size and form
–	71.	Stages continuous till
–	709	19th fruit has reached typical form and size

Table 8
Fruit ripening process.

2-digit	3-digit code	Description
81	801	10% of fruits show typical fully ripe, yellow color
82	802	20% of fruits show typical fully ripe, orange color
83	803	30% of fruits show typical fully ripe, orange color
84	804	40% of fruits show typical fully ripe, orange color
85	805	50% of fruits show typical fully ripe, orange color
86	806	60% of fruits show typical fully ripe, orange color
87	807	70% of fruits show typical fully ripe, orange color
88	808	80% of fruits show typical fully ripe, orange color
89	809	Fully ripe: fruits have typical fully ripe, orange color (Fig. 1L, M)

1995); however, later stages, such as stage (2), differ markedly in cape gooseberry due to the occurrence of the first bifurcation. This bifurcation is the start point for subsequent sympodyal growth. The sympodium is the linear series of shoots, each new distal shoot unit developing from the axillary bud situated on the previous shoot unit (Keller, 2004). The first bifurcation also marks the beginning of floral development. Thus, the first floral bud emerges on the first bifurcation. This event signifies the transition from vegetative to reproductive stage. Initiation in temperate herbaceous plants, such as *Arabidopsis*, has four pathways: venalization, photoperiod, autonomous floral initiation, and gibberellins that lead to floral formation (Bernier et al., 1993; Kinet et al., 1993; Boss et al., 2004; Corbesier and Coupland, 2005). Floral induction and subsequent initiation are linked events in some woody tropical plants (Ramírez and Davenport, 2010). Although, floral induction in cape gooseberry has been linked to photoperiod being a quantitative short day plant (Heinze and Midash, 1991), there is no substantial evidence to support this hypothesis because these authors failed to investigate the impacts of shorter photoperiods, such as 6 h and longer ones 10, 12 and 14, on floral induction.

Cape gooseberry fruiting occurs after fully expanded and mature flowers have been pollinated. After the first flower sets fruit, floral bud emergence continues to occur at each of the nodes at the reproductive parts of the plant. It is, thus, common to observe fully open flowers and early flowers as well as immature and mature fruits on the same plant. Moreover, as fruit develop on the plant's branches collectively, they tend to mature nearly synchronously, and thus, the plant produces great amounts of fruit during harvest as evidenced by our observation in our experiments and in field conditions (Table 8).

4. Conclusion

Cape gooseberry phenology is similar to the early developmental stages of other solanaceous fruits during germination (principle stage 0) and leaf development (principle stage 1). The principal stage 2: formation of side shoots, marks the beginning of a differentiating event, onset of the first bifurcation. At the first bifurcation, the first floral bud is initiated and marks the transition from vegetative to reproductive phase (principle stage 3). Subsequent bifurcations form a branched pattern. Each branch forms along the growing bifurcating stem. Initially, one flower and related leaves emerge at each bifurcating point. As the plant's stature increases, simultaneous stages occur. These are: the formation of side shoots (principle stage 2), inflorescence emergence (principle stage 5), flowering (principle stage 6), development of fruit (principle stage 7) and ripening of fruit and seed (principle stage 8). Fruit ripening occurs nearly synchronously with the calyx. The proposed scale is essential for proper management practices to be applied during the plant's life cycle.

Acknowledgements

Special thanks to L. Marien for her valuable support in this investigation.

References

Bernal, J.A., 1991. Agronomic aspects of the cultivation of the uchuva, *Physalis peruviana*, on the high plateau of the Colombian departments of Cundinamarca and Boyacá. In: Hawkes, J.G., Lester, R.N., Nee, M., Estrada, N. (Eds.), Solanaceae III: Taxonomy, Chemistry, Evolution. Royal Gardens Kew and Linnean Society of London, UK, pp. 459–460.
Bernier, G., Havelange, A., Houssa, C., Petitjean, A., Lejeune, P., 1993. Physiological signals that induce flowering. Plant Cell 5, 1147–1155.
Boss, P.K., Bastow, R.M., Milne, J.S., Dean, C., 2004. Multiple pathways in the decision to flower: enabling, promoting, and resetting. Plant Cell 16, S18–S31.
Corbesier, L., Coupland, G., 2005. Photoperiodic flowering of Arabidopsis: integrating genetic and physiological approaches to characterization of the floral stimulus. Plant. Cell Environment 28, 54–66.
Feller, C., Bleiholder, H., Buhr, L., Hack, H., Hess, M., Klose, R., Meier, U., Stauss, R., Van den Boom, T., Weber, E., 1995. Phänologische Entwicklungsstadien von Gemüsepflanzen: II. Fruchtgemüse und Hülsenfrüchte. Nachrichtenbl. Deut. Pflanzenschutzd. 47, 217–232.
Fischer, G., 2000. Crecimiento y desarrollo. In: Flórez, V.J., Fischer, G., Sora, A.D. (Eds.), Producción, poscosecha y exportación de la uchuva (*Physalis peruviana* L.). Unibiblos, Universidad Nacional de Colombia, Bogotá, pp. 9–26.
Fischer, G., 1995. Effect of root zone temperature and tropical altitude on the growth, development and fruit quality of cape gooseberry (*Physalis peruviana* L.). Ph.D. thesis. Universität Humboldt zu Berlin, Germany.
Fischer, G., 1989. Aspectos fisiológicos del desarrollo de la uchuva (*Physalis peruviana* L.). In: Proceedings of IV. Seminario Nacional Recursos Vegetales Promisorios UPTC, Tunja, Colombia, pp. 9–19.
Fischer, G., Ebert, G., Lüdders, P., 2007. Production, seeds and carbohydrate contents of cape gooseberry (*Physalis peruviana* L.) fruits grown at two contrasting Colombian altitudes. Journal of Applied Botany and Food Quality 81 (1), 29–35.
Galvis, J.A., Fischer, G., Gordillo, O.P., 2005. Cosecha y poscosecha de la uchuva. In: Fischer, G., Miranda, D., Piedrahita, W., Romero, J. (Eds.), Avances en cultivo, poscosecha y exportación de la uchuva (*Physalis peruviana* L.) en Colombia. Unibiblos, Universidad Nacional de Colombia, Bogotá, pp. 167–192.
Fischer, G., Herrera, A., Almanza, P.J., 2011. Cape gooseberry (*Physalis peruviana* L.). In: Yahia, E.M. (Ed.), Postharvest Biology and Technology of Tropical and Subtropical Fruits. Woodhead Publishing, Oxford, UK, pp. 374–396.
Fischer, G., Miranda, D., 2012. Uchuva (*Physalis peruviana* L.). In: Fischer, G. (Ed.), Manual para el cultivo de frutales en el trópico. Produmedios, Bogotá, pp. 851–873.
Gupta, S.K., Roy, S.K., 1981. The floral biology of cape gooseberry (*Physalis peruviana* Linn; Solanaceae, India). Indian Journal of Agricultural Sciences 51 (5), 353–355.
Heinze, W., Midash, M., 1991. Photoperiodische Reaktion von *Physalis peruviana* L. Gartenbauwiss 56 (6), 262–264.
Keller, R., 2004. Identification of Tropical Woody Plants in the Absence of Flowers, 2nd Edition. Birkhäuser Verlag, Berlin.
Kinet, J.-M., Lejeune, P., Bernier, G., 1993. Shoot-root interactions during floral transition: a possible role for cytokinins. Environmental and Experimental Botany 33, 459–469.
Legge, A.P., 1974. Notes on the history, cultivation and uses of *Physalis peruviana* L. Journal of the Royal Horticultural Society 99 (7), 310–314.
McCain, R., 1993. Goldenberry, passionfruit & white sapote: potencial for cool subtropical areas. In: Janick, J., Simon, J.E. (Eds.), New crops. Wiley, New York, pp. 479–486.
Meier, U., 1997. Growth Stages of Mono- and Dicotyledoneous Plants. Blackwell Science, New York.
National Research Council, 1989. Lost Crops of the Incas. National Academy Press, Washington, D.C, pp. 241–251.
Puente, L.A., Pinto-Muñoz, S.A., Castro, E.S., Cortés, M., 2011. Physalis peruviana Linnaeus, the multiple properties of a highly functional fruit: a review. Food Research International 44, 1733–1740.
Ramírez, F., Davenport, T.L., 2010. Mango (*Mangifera indica* L.) flowering physiology. Scientia Horticulturae 126 (2), 65–72.
Whitson, M., Manos, P.S., 2005. Untangling *Physalis* (Solanaceae) from the Physaloids: a two-gene phylogeny of the Physalinae. Systematic Botany Monographs 30, 216–230.